

General Information

The images listed below will assist you in preparing for the lab during open hours. Use these and compare them with the images provided in this document.

Literature

Available in the *McMinn's & Abrahams' Clinical Atlas of Human Anatomy*:

Chapter 1: Skull pg 9-11

Chapter 1: Neck pg 28 - 37

Chapter 1: Face pg 38 – 41

Chapter 1: Infratemporal fossa pg 42 – 44

Chapter 1: Cranial Fossa pg 59-61

Literature

Available in the Gray's Clinical dissector

Chapter 20: Neck pg 336- 349 Images: 20.19, 20.20, 20.21, 20.23, 20.26, 20.36, 20.37, 20.41, 20.53

Chapter 21: Face pg 359 – 362 Images: 21.17, 21.18, 21.19, 20.29

Chapter 22: Infratemporal fossa pg 364 – 375 Images: 22.1, 22.5, 22.6, 22.10, 22.12, 22.18, 22.19, 22.20, 22.21, 22.27, 22.28, 22.29, 22.33, 22.34

Chapter 23: Calvaria, dural venous sinuses, cranial nerves pg 380 – Images: 23.11, 23.16, 23.17, 23.43, 23.45, 23.49,

Chapter 27: Oral Cavity pg 426 – 435 Images: 27.2, 27.6, 27.8, 27.14, 27.18, 27.20, 27.22, 27.23

Cranial Cavity and Cranial Nerves₁



Discuss the boundaries of cranial fossa (anterior, middle and posterior) and contents of each.

Compare what is visible on the dry skull versus the skull lined with dura mater.

In the cranial cavity identify the:

- cribriform plate, crista galli, anterior and posterior clinoid processes, orbital canal, superior and inferior orbital fissure, optic foramen, ethmoid foramina, foramen ovale, foramen rotundum, foramen spinosum, internal acoustic meatus, jugular foramen, foramen magnum, hypoglossal canal, foramen lacerum and groove for the middle meningeal artery
- tentorium cerebelli, falx cerebri, falx cerebelli
- optic, oculomotor, trochlear, trigeminal, abducens, facial and vestibulocochlear, glossopharyngeal and vagus, spinal accessory and hypoglossal nerves
- ophthalmic, internal carotid and vertebral arteries

On the brain identify:

- Olfactory bulb and tract, optic, oculomotor, trochlear, trigeminal, abducens, facial, vestibulocochlear, glossopharyngeal, vagus and hypoglossal nerves
- Internal carotid, anterior cerebral, anterior communicating, posterior communicating, posterior cerebral, superior cerebellar, basilar and vertebral arteries

Clinical considerations:

Discuss the type of fibers found in each of the cranial nerves.

Discuss the foramen through which each cranial nerve exits the skull and the structures they supply.

Discuss the consequence of damage or a space occupying lesion near each of the following locations:

- Cavernous sinus Internal acoustic meatus Jugular foramen

Cranial Cavity and Cranial Nerves₂



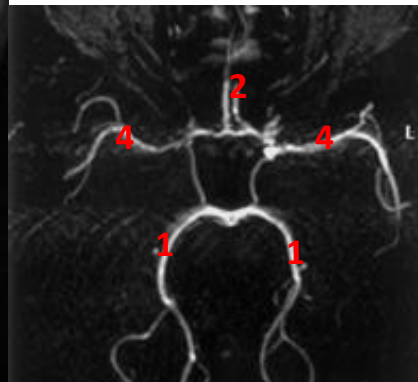
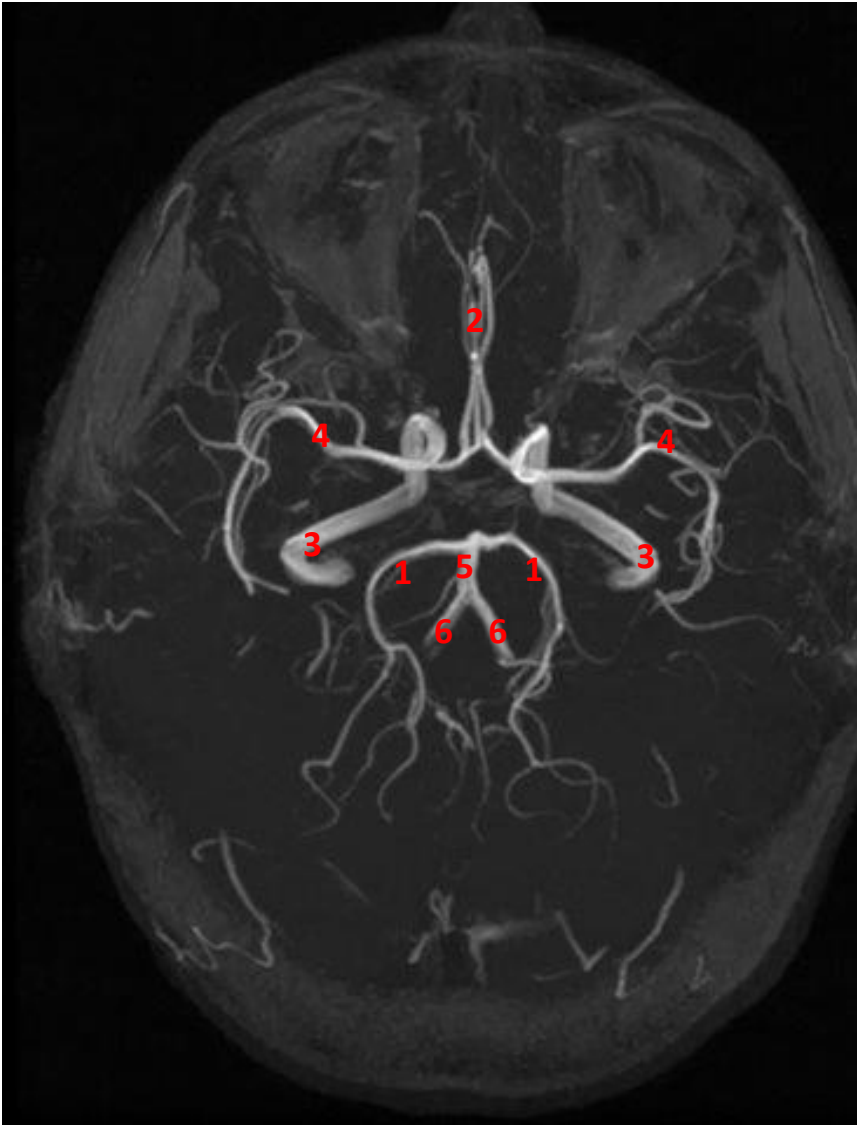
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Include the modality and the part of the body being viewed.

Identify the following arteries:

- Posterior cerebral
- Anterior cerebral
- Internal carotid
- Middle cerebral
- Basilar
- Vertebral

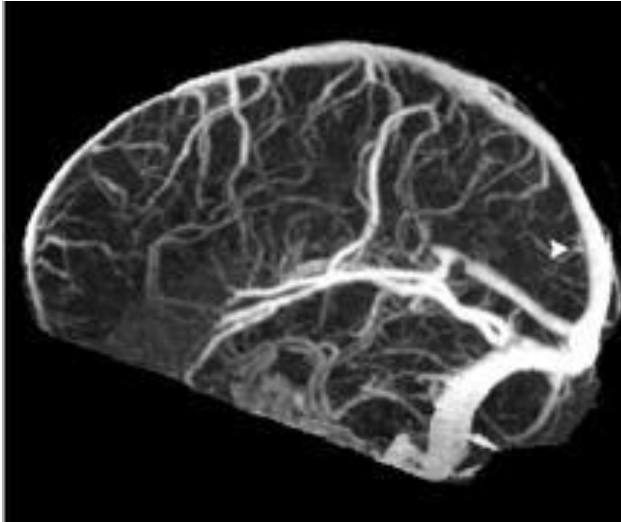


Cranial Cavity and Cranial Nerves₃



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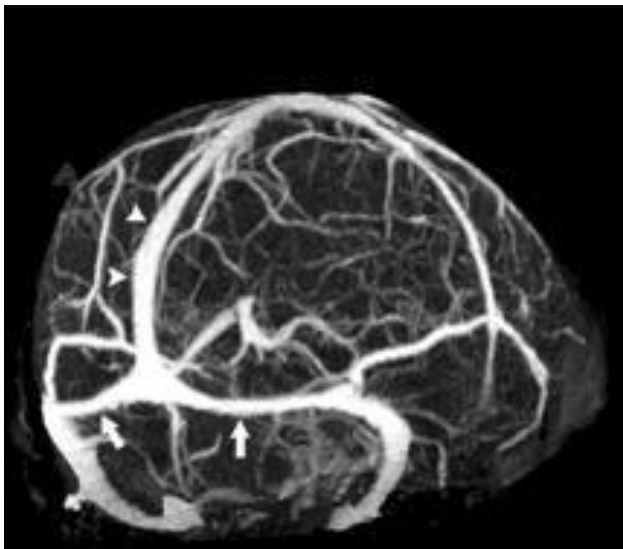
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What is the investigative modality seen here?

Identify the following sinuses:

- * Superior sagittal
- * Transverse
- * Sigmoid
- * Straight
- * Confluence



Are these sinuses veins? Discuss.

Discuss the sites of production of CSF and the route it takes to its final destination.

Where do these sinuses eventually drain to?

Triangles of the Neck₁



Discuss the boundaries of the anterior and posterior triangles of the neck.

Identify the:

- sternocleidomastoid, omohyoid, sternohyoid, sternothyroid, thyrohyoid, digastric, stylohyoid, anterior, middle and posterior scalene muscles
- submental, submandibular, carotid, muscular, suboccipital and supraclavicular triangles
- discuss how the muscles contribute to the subdivisions of the anterior triangle and the significance of these subdivisions

Identify the:

- subclavian, common carotid, external carotid, internal carotid, lingual, facial, transverse cervical, inferior thyroid, superior thyroid arteries
- hypoglossal, vagus, internal laryngeal, C1, ansa-cervicalis and its branches (*transverse cervical, great auricular, lesser occipital, supraclavicular*), spinal accessory, phrenic nerves and trunks of brachial plexus
- internal jugular, *external jugular*, subclavian veins
- submandibular and thyroid glands

Clinical considerations:

Discuss the clinical significance of the junction between the internal jugular and subclavian veins.

- Discuss the placement of a central venous line
- Which structure is related to the left and right sides respectively?

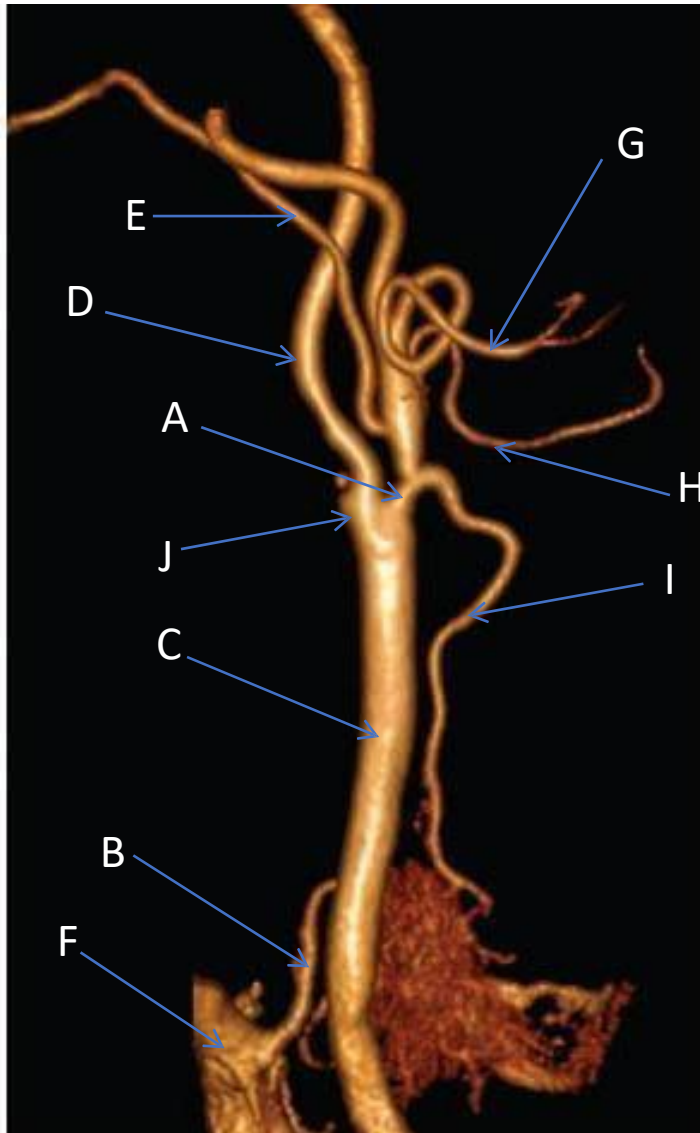
Discuss the location and significance of the carotid sinus and body.

Discuss what is meant with “carefree” and “careful” triangles.

Triangles of the Neck₂



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What is the investigative modality shown here?

Identify A - J:

What are the terminal branches of the artery labeled A?

IT, PT Fossa



From the lateral view (IT) Identify:

- mandible, zygomatic bone, temporal bone and maxilla
- medial and *lateral pterygoid*, temporal and masseter muscles
- mandibular, lingual, *chorda tympani*, inferior alveolar, mylohyoid, auriculotemporal and buccal nerves
- maxillary, middle meningeal, *posterior superior alveolar*, inferior alveolar and deep temporal arteries

From the medial view (PT) Identify:

- lateral pterygoid plate, maxilla, pterygomaxillary fissure
- foramen rotundum and infraorbital fissure
- *pterygopalatine ganglion*, maxillary, infraorbital, descending palatine nerves and nerve of the pterygoid canal
- sphenopalatine, infraorbital, descending palatine arteries

Clinical considerations:

Discuss the origin and course of the parasympathetic fibers that synapse in and the sympathetic fibers that pass through the PT ganglion.

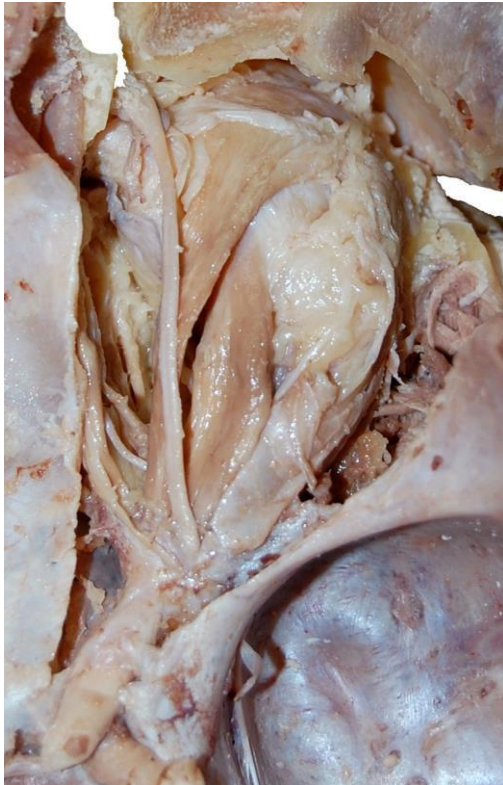
Discuss the distribution of the PT ganglion.

Orbit₁



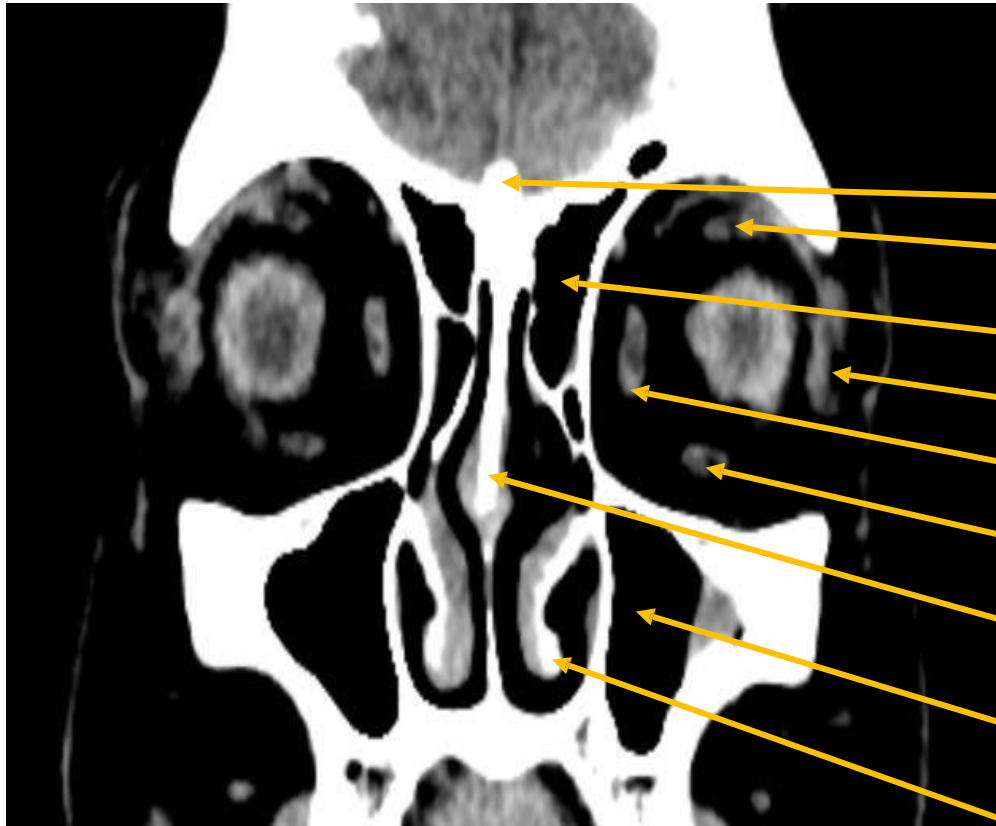
Identify:

- frontal, maxilla, lacrimal, palatine, zygomatic, ethmoid and sphenoid bones
- superior, *medial*, lateral and inferior rectus, superior oblique, *inferior oblique*, levator palpebrae superioris muscles
- oculomotor, trochlear, optic, anterior ethmoidal and abducens nerves
- ophthalmic artery
- lacrimal gland



What is the investigative modality shown here?

Identify A-I



A

B

C

D

E

F

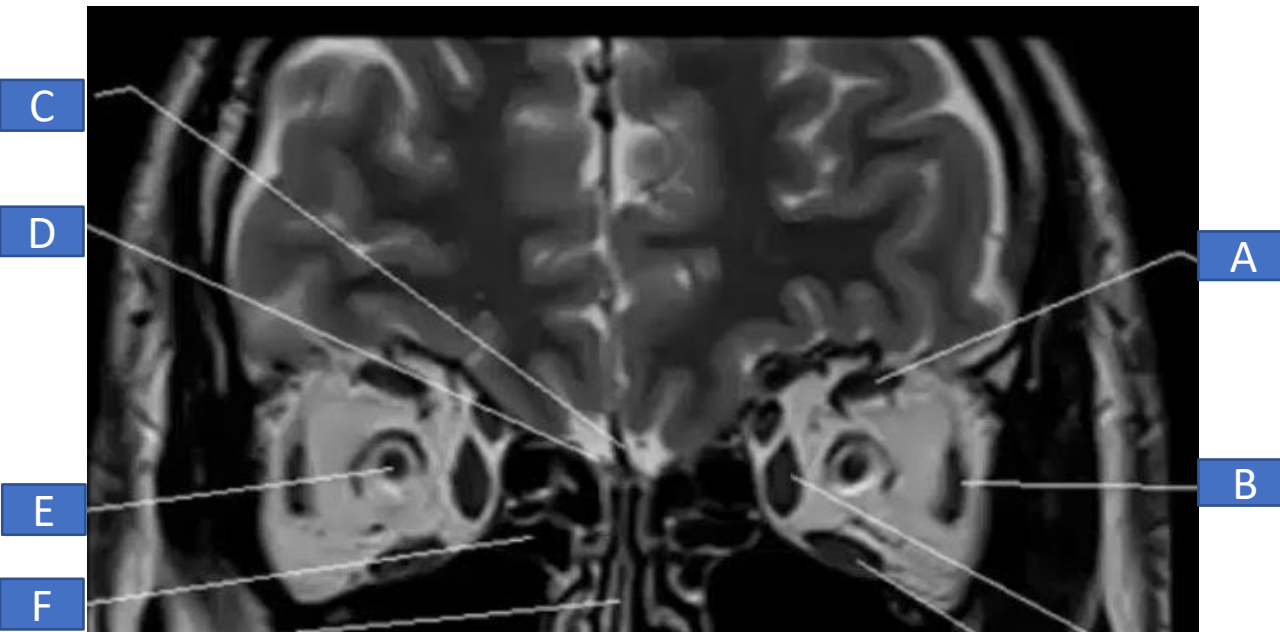
G

H

I

What is the investigative modality shown here?

Identify A to F:



Nasal Cavity, Oral Cavity, Pharynx and Larynx₁



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Identify:

- nasal bones, nasal bony and cartilaginous septum, choanae, hard palate, ethmoid bone, nares, concha, meati, openings for drainage of the paranasal sinuses and nasolacrimal duct
- alveolar processes of maxilla and mandible, soft palate

Discuss the relationship between the oral and nasal cavities; how is food prevented from entering the nasal cavity during swallowing?

Identify (pharynx):

- nasopharynx, oropharynx, laryngopharynx
- superior-, middle-, inferior constrictors, stylopharyngeus, salpingopharyngeus, tensor veli palatini, levator veli palatini muscles
- glossopharyngeal nerve, vagus nerve, and cervical sympathetic chain
- torus tubarius, opening of the pharyngotympanic tube, piriform recess

Nasal Cavity, Oral Cavity, Pharynx and Larynx₂



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Identify (larynx):

- thyroid, cricoid, arytenoid, epiglottis cartilages
- transverse arytenoid and posterior cricoarytenoid muscles
- vocal and vestibular folds
- vestibule, ventricle, infraglottic spaces

Identify (Oral Cavity)

- glossopharyngeal, lingual, hypoglossal nerves
- lingual artery
- submandibular gland and duct, sublingual and parotid glands – Discuss where the openings of the ducts of these glands would be located
- palatoglossus, styloglossus, hyoglossus, genioglossus, mylohyoid, geniohyoid muscles

Clinical considerations:

Discuss the innervation to the tongue.

Discuss the relationship between the lingual nerve and submandibular duct.

Discuss and explain the actions of the muscles of the soft palate during swallowing

Discuss the drainage of the paranasal sinuses and nasolacrimal duct into the nasal cavity

Discuss the blood supply of the nasal cavity

- Discuss the anastomosis at Kiesselbach's area and why it may be prone to epistaxis ?

Discuss the innervation of the hard and soft palate

- Identify the image modality used here and name the plane of the image taken.

- Identify and name the structures indicated by the numbered arrows.

Include in your discussion:

- the structure that attaches to (3)
- the bony origins of structures labeled (5) and (6)
- What spaces are 1, 2 & 4

Observe and discuss the abnormality seen at the solid red arrow. Include in your discussion, possible consequences of this abnormality.

